Original Article — DRAFT: every red [TBD] must be filled from experiment outputs before submission

BrainHybridNet: A Lightweight Hybrid CNN–Transformer with Self-Supervised Pre-training and Uncertainty Estimation for Joint Brain Tumor Segmentation, Detection and Grading from Multi-Modal MRI

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Abstract

Purpose: Automated analysis of brain tumors in multi-modal magnetic resonance imaging (MRI) usually relies on separate models for segmentation, detection and grading, many of which are large and report no measure of confidence.

Methods: We propose BrainHybridNet, a lightweight hybrid convolutional neural network (CNN)–vision transformer that segments the whole tumor, tumor core and enhancing tumor in 3D, detects tumor presence and predicts low- versus high-grade glioma in a single forward pass. A sequence-attention block and a two-stage CNN stem provide high-resolution features and skip connections; a four-layer transformer models long-range context. The encoder is pre-trained with 3D masked image modeling, Monte Carlo dropout provides voxel-wise epistemic uncertainty, and transformer attention provides spatial explanations. All baselines (3D U-Net, SegResNet, UNETR, SwinUNETR and nnU-Net) are re-trained on identical subject-level splits of BraTS 2021 (1,251 subjects).

Results: On the held-out test set BrainHybridNet reaches Dice scores of **[TBD]** (whole tumor), **[TBD]** (tumor core) and **[TBD]** (enhancing tumor) with **[TBD]** million parameters and **[TBD]** ms per 128^3 crop. **[TBD: grading, calibration and external-validation results]**.

Conclusion: **[TBD: one sentence on the measured accuracy–efficiency–calibration trade-off]**.

Keywords Brain tumor segmentation · Multi-modal MRI · Hybrid CNN–transformer · Masked image modeling · Uncertainty estimation · Glioma grading

1 Introduction

Brain tumors are among the most aggressive forms of cancer: glioblastoma (WHO grade 4) has a median overall survival of 14.6 months under standard-of-care treatment [1]. About 308,000 new cases of brain and central nervous system tumors were estimated worldwide in 2020 [2]. Accurate delineation of the tumor sub-regions—whole tumor (WT), tumor core (TC) and enhancing tumor (ET)—is a prerequisite for surgical planning, radiotherapy targeting and response monitoring [3].

Multi-modal MRI is the standard for non-invasive brain tumor characterization. T1-weighted images show anatomy, contrast-enhanced T1 (T1ce) shows the enhancing tumor, T2-weighted images show peritumoral edema and FLAIR is sensitive to infiltrative tumor [4]. Manual delineation takes a long time per case and varies between raters (Dice agreement of 74–85%) [3], which motivates automated methods.

CNNs, particularly U-Net and its 3D variants, have dominated medical image segmentation since 2015 [5, 6]. Their local receptive fields limit the modeling of long-range spatial dependencies, which matter for large, irregular tumors [7]. Transformers [8] attend to all positions at once, but pure vision transformers need large training sets and are expensive for 3D volumes [7].

Existing methods leave several gaps:

1. **Single-task designs:** segmentation and grading are usually solved by separate models [9].
2. **Data requirements:** transformer models need many labeled scans [10].
3. **Missing uncertainty:** most segmentation methods report point predictions only [11].
4. **Limited interpretability:** clinical use requires explanations of model decisions [12].
5. **Computational cost:** models with tens of millions of parameters are hard to deploy on standard workstations [13].

We propose BrainHybridNet, which combines a CNN stem (local features) and a transformer bottleneck (global context) with self-supervised pre-training, uncertainty estimation and multi-task heads.

1.1 Contributions

1. **Lightweight hybrid architecture.** A ResNet-style CNN stem extracts local features and provides high-resolution skip connections; a vision-transformer bottleneck captures volumetric long-range dependencies with **[TBD]** million parameters.
2. **3D masked image modeling on multi-modal MRI.** Following MAE [14] and SimMIM [15], 75% of 3D patches are masked and reconstructed before fine-tuning. Unlike self pre-training of a plain ViT on the target data [16], masking is applied before the CNN stem of a hybrid encoder, and its effect on accuracy and label efficiency is measured (Section 6).
3. **Joint multi-task learning.** One model outputs tumor presence, WT/TC/ET masks and an LGG/HGG grade.
4. **Uncertainty via Monte Carlo dropout.** T stochastic passes give a voxel-wise epistemic uncertainty map; we test whether it tracks segmentation error and how it affects calibration.
5. **Attention-based explanations.** Attention weights from the last transformer block give a saliency map, evaluated quantitatively against reference masks.
6. **Reproducible comparison.** All baselines are re-trained on identical splits and preprocessing; results include confidence intervals and paired significance tests; code, splits and weights are released.

Section 2 reviews related work, Section 3 describes BrainHybridNet, Section 4 the experimental setup, Section 5 the results, Section 6 the ablations, Section 7 the discussion and Section 8 concludes.

2 Related work

2.1 CNN-based brain tumor segmentation

U-Net [5] established encoder–decoder segmentation with skip connections, and 3D U-Net [6] extended it to volumes. SegResNet [17], winner of BraTS 2018, uses a residual encoder with variational auto-encoder regularization. MONAI [18] provides optimized implementations of these models. nnU-Net [19] configures U-Net pipelines automatically and remains a strong baseline.

2.2 Transformer-based methods

The vision transformer [7] matches CNNs when trained at scale. TransUNet [10] uses a transformer bottleneck in a U-Net. The Swin transformer [20] computes attention in shifted windows; SwinUNETR [21] applies it to 3D brain tumor segmentation, UNETR [22] uses a pure transformer encoder, and TransBTS [23] combines a 3D CNN encoder with a transformer for BraTS. These models need substantial GPU memory.

2.3 Hybrid CNN–transformer architectures

MedT [24] uses gated axial attention, UCTransNet [25] replaces skip connections with channel transformers and D-Former [26] uses dilated attention for 3D segmentation. BrainHybridNet combines an explicit CNN stem, a small transformer with learned 3D positional embeddings and a U-Net decoder in one multi-task model.

2.4 Self-supervised learning in medical imaging

Contrastive methods such as SimCLR [27] and DINO [28] have been adapted to medical images [29]. Masked autoencoders [14] learn by reconstructing a large fraction (75%) of masked patches. Zhou et al. [16] showed that MAE self pre-training on the target dataset improves segmentation, including BraTS. Swin UNETR pre-training [21] combines several proxy tasks on CT.

2.5 Uncertainty estimation

Uncertainty in deep learning has been approached with Bayesian methods [30], deep ensembles [31] and Monte Carlo dropout [11]. Nair et al. [32] found that uncertain regions in multiple-sclerosis lesion segmentation coincide with errors, and Roy et al. [33] used Monte Carlo sampling for segmentation quality control.

2.6 Glioma grading

The 2021 WHO classification combines histological and molecular markers [34]. MRI-based grading and molecular prediction have been studied with radiomics [35] and CNNs [36], typically as pipelines separate from segmentation.

3 Method

3.1 Overview

Fig. 1 shows BrainHybridNet. It consists of (1) a sequence-attention block and CNN stem, (2) a transformer encoder on the CNN features, (3) a U-Net decoder fusing the CNN skip features and (4) task heads for segmentation, detection and grading. An optional masked-image-modeling stage pre-trains the encoder.

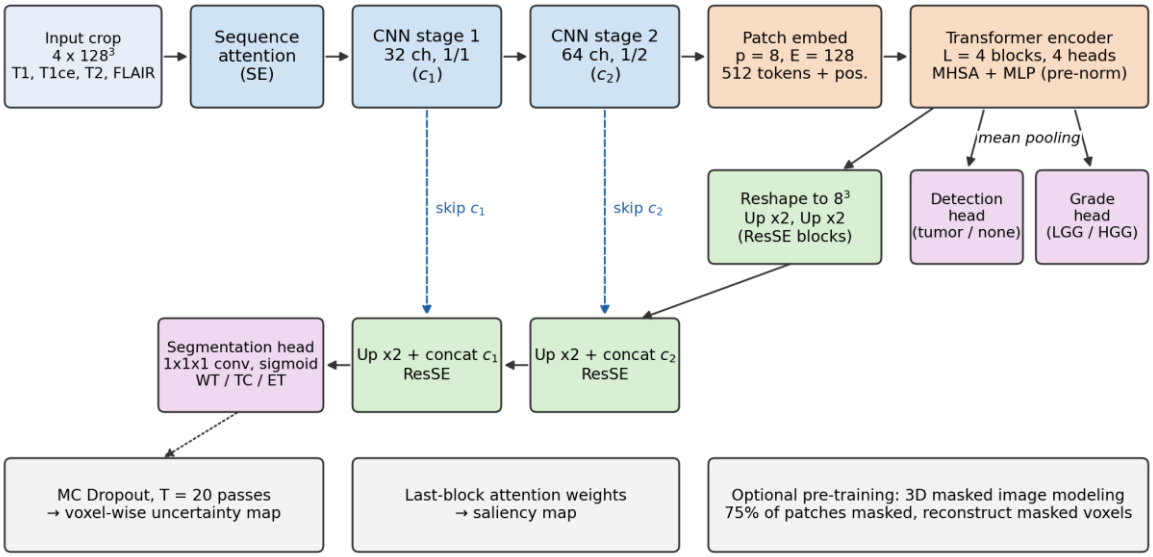


Fig. 1 BrainHybridNet. A squeeze-and-excitation block weights the four MRI sequences; a two-stage CNN stem produces skip features c_1 and c_2; 512 tokens from 8^3 patches of c_2 pass through four transformer blocks. Mean-pooled tokens feed the detection and grade heads; the decoder upsamples the token grid and fuses c_2 and c_1 before the WT/TC/ET segmentation head. Monte Carlo dropout gives an uncertainty map and last-block attention a saliency map.

3.2 Input preprocessing

The input is $\mathbf{X} \in \mathbb{R}^{B\times 4\times D\times H\times W}$. BraTS 2021 volumes are skull-stripped, co-registered to the SRI24 atlas and resampled to 1 mm isotropic resolution [37], so no further registration is applied. Each sequence is z-score normalized over non-zero (brain) voxels and cropped to the brain bounding box. Training uses random 128^3 crops; with probability 0.66 a crop is centered on a random tumor voxel. Inference on whole volumes uses sliding windows of the same size with 50% overlap and Gaussian weighting.

3.3 Sequence attention and CNN stem

A squeeze-and-excitation block re-weights the four sequences. The CNN stem has two stages without max-pooling:

$$c_1 = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv3D}_{3,1}(\mathbf{X}))) \in \mathbb{R}^{B\times 32\times D\times H\times W} \tag{1}$$

$$c_2 = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv3D}_{3,2}(c_1))) \in \mathbb{R}^{B\times 64\times D/2\times H/2\times W/2} \tag{2}$$

where the subscripts give kernel size and stride. c_1 and c_2 are the decoder's skip connections.

3.4 Transformer encoder

A strided convolution splits c_2 into non-overlapping 8^3 patches, giving $N = (D/16)(H/16)(W/16)$ tokens of dimension $E = 128$ ($N = 512$ for a 128^3 crop), to which learned positional embeddings $\mathbf{P} \in \mathbb{R}^{1\times N\times E}$ are added:

$$z_0 = \mathrm{Flatten}(\mathrm{Conv3D}_{8,8}(c_2))^{\top} + \mathbf{P} \tag{3}$$

$L = 4$ pre-norm transformer blocks [8, 38] with 4 heads, MLP ratio 4, GELU [39] and dropout 0.1 follow:

$$z'_l = z_l + \mathrm{MHSA}(\mathrm{LN}(z_l)), \quad z_{l+1} = z'_l + \mathrm{MLP}(\mathrm{LN}(z'_l)) \tag{4}$$

and a final layer normalization gives $z_L \in \mathbb{R}^{B\times N\times E}$.

3.5 Masked image modeling pre-training

Pre-training follows SimMIM [15] in 3D. A fraction $\rho = 0.75$ of token positions is masked at random. The corresponding 16^3 input regions are zeroed before the CNN stem, so masked content cannot leak through the convolutions, and masked tokens are replaced by a learned mask embedding. A linear head maps each output token to the 4×16^3 voxels of its region. The loss is the mean squared error to per-patch normalized targets over the masked set $\mathcal{M}$:

$$\mathcal{L}_{\text{MIM}} = (1/|\mathcal{M}|) \, \Sigma_{i \in \mathcal{M}} \, \| \hat{x}_i - \text{norm}(x_i) \|^2 / (4 \cdot 16^3) \tag{5}$$

Pre-training uses only the training folds of each cross-validation split; the reconstruction head is discarded before fine-tuning.

3.6 Decoder

The tokens are reshaped to $z_{\text{vol}} \in \mathbb{R}^{B \times E \times D/16 \times H/16 \times W/16}$ and passed through 3D dropout (0.1). Three upsampling stages (stride-2 transposed convolution followed by a residual squeeze-and-excitation block, ResSE) restore the resolution of c_2, which is concatenated before the third ResSE block; a fourth stage upsamples to full resolution and fuses c_1:

$$u = \text{ResSE}([\text{Up}(\text{Up}(\text{Up}(z_{\text{vol}}))), c_2]) \in \mathbb{R}^{B \times 64 \times D/2 \times H/2 \times W/2} \tag{6}$$

$$d = \text{ResSE}([\text{Up}(u), c_1]) \in \mathbb{R}^{B \times 32 \times D \times H \times W} \tag{7}$$

3.7 Task heads

Segmentation: a 1×1×1 convolution gives three channels (WT, TC, ET), each with a sigmoid. **Detection:** mean-pooled tokens $\bar{z} = \text{mean}(z_L)$ feed Linear(E,128) → ReLU → Dropout(0.1) → Linear(128,2). BraTS has no tumor-free subjects, so detection is trained at crop level: a crop is positive if it contains at least 100 whole-tumor voxels. **Grading:** Linear(E,64) → ReLU → Dropout(0.1) → Linear(64,2) predicts LGG versus HGG; subjects without a grade label are ignored by this loss.

3.8 Uncertainty estimation

At inference, batch normalization stays in evaluation mode and dropout stays active [11]. $T = 20$ stochastic sliding-window passes give

$$\mu = (1/T) \, \Sigma_{t=1}^{T} \, \sigma(\hat{Y}^{(t)}), \quad \sigma^2 = (1/T) \, \Sigma_{t=1}^{T} \, (\sigma(\hat{Y}^{(t)}) - \mu)^2 \tag{8}$$

where σ^2 is the voxel-wise epistemic uncertainty map.

3.9 Training objective and optimization

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{seg}} + \lambda_2 \mathcal{L}_{\text{det}} + \lambda_3 \mathcal{L}_{\text{grade}}, \quad \lambda_1 = 0.6, \lambda_2 = \lambda_3 = 0.2 \tag{9}$$

$\mathcal{L}_{\text{det}}$ and $\mathcal{L}_{\text{grade}}$ are cross-entropy losses. $\mathcal{L}_{\text{seg}}$ is the Tversky loss [40], averaged over the three region channels:

$$\text{TL} = 1 - (\text{TP} + \varepsilon) / (\text{TP} + \alpha \cdot \text{FP} + \beta \cdot \text{FN} + \varepsilon), \quad \alpha = 0.3, \beta = 0.7 \tag{10}$$

which penalizes false negatives more than false positives. The loss weights were fixed before training; their sensitivity is reported in Section 6.

Optimization uses AdamW [41] (learning rate 10^{-4}, weight decay 10^{-4}, gradient clipping at norm 1.0), 10 warm-up epochs followed by cosine decay [42], 300 fine-tuning epochs and 100 pre-training epochs (learning rate 1.5×10^{-4}), batch size 2 with mixed precision. Augmentation: random flips along each axis ($p = 0.5$), per-sequence intensity scaling (0.9–1.1) and shift (±0.1), Gaussian noise ($\sigma = 0.01$, $p = 0.2$). The checkpoint with the highest mean validation Dice of each fold is kept.

4 Experimental setup

4.1 Data

BraTS 2021 [37]: 1,251 subjects with expert segmentations and four co-registered sequences (240 × 240 × 155 voxels at 1 mm^3). Labels are background (0), necrotic core (1), edema (2) and enhancing tumor (4); WT = 1 ∪ 2 ∪ 4, TC = 1 ∪ 4, ET = 4.

Splits. A fixed subject-level hold-out test set of 15% of subjects (**[TBD: n]** subjects) is used once, for the final results. The remaining subjects form 5 cross-validation folds shared by every model, baseline and ablation; the split file is released with the code.

Grading data. BraTS 2021 has no histological grades. Grading is trained and evaluated on UCSF-PDGM [43] (WHO grade for preoperative diffuse gliomas) with its own subject-level split **[TBD: confirm]**, which also serves as an external segmentation test set without retraining.

4.2 Metrics

Segmentation: Dice and 95th-percentile Hausdorff distance (HD95, mm) per region, sensitivity and specificity. Following BraTS, Dice = 1 and HD95 = 0 when prediction and reference are both empty, and HD95 = 373.13 mm when only one is empty. **Detection and grading:** AUC, accuracy, balanced accuracy, sensitivity, specificity. **Calibration and uncertainty:** voxel-level expected calibration error (15 bins) and reliability diagrams [44]; Spearman correlation between case-level mean uncertainty and 1 − Dice. **Efficiency:** parameters, multiply–accumulate operations, median latency per 128^3 crop and peak GPU memory on **[TBD: GPU]**. **Statistics:** mean ± standard deviation over cases and 95% bootstrap confidence intervals (10,000 resamples); paired Wilcoxon signed-rank tests against the strongest baseline with Holm correction. Reporting follows the CLAIM checklist [45] (Supplementary Material).

4.3 Baselines and implementation

All baselines—3D U-Net [6], SegResNet [17], UNETR [22], SwinUNETR [21] (MONAI implementations [18]) and nnU-Net v2 [19] (default pipeline)—are re-trained by us on the same folds with the same crops, preprocessing, augmentation, loss and schedule; no numbers are copied from other publications. Models are implemented in PyTorch [46] with PyTorch Lightning [47]. Experiments ran on **[TBD: hardware]**; pre-training took **[TBD]** h and fine-tuning **[TBD]** h per fold.

5 Results

5.1 Segmentation

Table 1 lists the test-set results and Table 2 the cross-validation results. **[TBD: describe the measured differences only]**.

Table 1 Segmentation on the BraTS 2021 hold-out test set (**[TBD]** subjects), mean ± SD. Best value per column in bold

Method	Dice ↑			HD95 (mm) ↓			Params (M)	GMACs	Latency (ms)
	WT	TC	ET	WT	TC	ET			
3D U-Net [6]	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
SegResNet [17]	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
UNETR [22]	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
SwinUNETR [21]	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**

Method	Dice ↑			HD95 (mm) ↓			Params (M)	GMACs	Latency (ms)
	WT	TC	ET	WT	TC	ET			
nnU-Net v2 [19]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
BrainHybridNet (ours)	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

Significance versus the strongest baseline (Wilcoxon, Holm-corrected): [TBD].

Table 2 Five-fold cross-validation, mean ± SD of fold means (Dice)

Method	WT	TC	ET
Best baseline ([TBD])	[TBD]	[TBD]	[TBD]
BrainHybridNet	[TBD]	[TBD]	[TBD]

5.2 Detection and grading

Table 3 Crop-level tumor detection (BraTS test set) and LGG/HGG grading (UCSF-PDGM test split)

	Det. AUC	Sens.	Spec.	Grade AUC	Grade bal. acc.
BrainHybridNet	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

5.3 Calibration and uncertainty

[TBD: voxel ECE with and without Monte Carlo dropout and for the baselines; reliability diagram (Fig. 2); Spearman ρ between case-level uncertainty and 1 − Dice with p-value].

5.4 External validation

[TBD: segmentation on UCSF-PDGM without retraining].

5.5 Qualitative results

[TBD: Fig. 3 – predictions, references, uncertainty and attention maps for representative successes and failures].

6 Ablation study

Table 4 Ablations, 5-fold cross-validation on the development set (Dice)

Configuration	WT	TC	ET	Params (M)
CNN only (no transformer)	[TBD]	[TBD]	[TBD]	[TBD]
Transformer path only (no CNN skips)	[TBD]	[TBD]	[TBD]	[TBD]
No sequence attention	[TBD]	[TBD]	[TBD]	[TBD]
Segmentation only (no multi-task)	[TBD]	[TBD]	[TBD]	[TBD]
No self-supervised pre-training	[TBD]	[TBD]	[TBD]	[TBD]
Full BrainHybridNet	[TBD]	[TBD]	[TBD]	[TBD]

Configuration files: abl_cnn_only, abl_no_skips, abl_no_se, abl_no_multitask, hybrid (with and without pre-training).

[TBD: masking ratio (0.25/0.5/0.75/0.9), number of transformer layers (2/4/6/8) and loss-weight sensitivity, three seeds each].

7 Discussion

7.1 Clinical implications

A model that segments, detects and grades in one pass could shorten reading time if its accuracy matches dedicated models; the uncertainty map is intended to direct review to ambiguous regions, which requires the correlation shown in Section 5. **[TBD: interpret the measured results]**.

7.2 Pre-training and data efficiency

[TBD: effect of masked image modeling, including reduced-label experiments with 10% and 25% of labeled subjects].

7.3 Explainability

[TBD: overlap or pointing-game evaluation of attention maps against reference masks].

7.4 Limitations

1. External validation is limited to **[TBD]**.
2. WHO CNS5 classification uses IDH and other molecular markers [34]; the grading head uses imaging only.
3. Pre-training uses only BraTS training folds; larger unlabeled corpora may help.
4. BraTS has no tumor-free subjects, so detection is evaluated at crop level, not on healthy scans.
5. BraTS labels have known inter-rater variability [3].

7.5 Future work

Federated training across hospitals, longitudinal response assessment, pediatric tumors, robustness to missing sequences, fusion with molecular data and a prospective reader study.

8 Conclusion

We presented BrainHybridNet, a lightweight hybrid CNN–transformer for joint brain tumor segmentation, detection and grading from multi-modal MRI, with masked image modeling pre-training and Monte Carlo dropout uncertainty. **[TBD: one sentence summarizing measured accuracy, efficiency and calibration relative to the baselines]**. Code, configurations, splits and weights are released to reproduce every reported number.

Supplementary information

CLAIM checklist; per-fold results; additional qualitative examples. **[TBD: confirm contents]**.

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- **Competing interests:** The authors declare no competing interests.

- **Ethics approval and consent to participate:** Not applicable. This study uses publicly available, de-identified datasets (BraTS 2021, UCSF-PDGM) under their data-use terms; no new data were collected.
- **Consent for publication:** Not applicable.
- **Data availability:** BraTS 2021 is available at https://www.synapse.org/#!Synapse:syn27046444. UCSF-PDGM is available through The Cancer Imaging Archive under its data-use agreement [43].
- **Materials availability:** Not applicable.
- **Code availability:** Code, configuration files, data splits and trained weights are available at https://github.com/Debobrato06/BrainTumorDetection **[TBD: release tag / DOI, e.g. Zenodo]**.
- **Author contribution:** D.B.: conceptualization, methodology, software, validation, writing – original draft. **[TBD: co-author initials]**: writing – review and editing, visualization.

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